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Lab 7.

1.

Process	AT	BT	CT	TAT	WT
P_1	0	6	6	6	0
P_2	2	8	14	12	4
P_3	4	7	21	17	10
P_4	6	3	24	18	15

Grant chart

P_1	P_2	P_3	P_4

0 6 14 21 24

$TAT \; \frac{\Sigma T_i}{n} = \frac{6+12+17+18}{4} = 13.25$ unit.

avg $WT = \frac{0+4+10+15}{4} = 7.25$ unit.

2 SJF.

Proces	AT	BT	CT	TAT	WT
P_1	1	5	6	5	0
P_2	3	2	8	5	3
P_3	5	9	21	16	7
P_4	6	4	12	6	2

Ideal	P_1	P_2	P_4	P_3

0 1 6 8 12 21.

avg tat $= \frac{5+5+16+6}{4} = 8$ unit

avg WT $= \frac{0+3+7+2}{4} = 3$ unit.

Scanned with CamScanner

Ta

3.

P1	P2	P2	P2	P2	P4	P4	P3	P1	
0	1	2	3	4	5	6	7	14	22.

Process	AT	BT	CT	TAT	WT
P_1	0	9	22	22	13
P_2	1	4	5	4	0
P_3	2	7	14	12	5
P_4	3	2	7	4	2.

avg tat = 10.5 unit
avg wt = 5 unit.

4.

Process	AT	BT	CT	TAT	WT.
P_1	0	10	26	26	16
P_2	1	5	17	16	11
P_3	2	8	25	23	15
P_4	3	3	12	9	6.

P1	P2	P2	P4	P1	P2	P3	P1	P3	P1	
0	3	6	9	12	15	17	20	23	25	26

avg tat = 18.5 unit.

5.

Process	AT	BT	CT	TAT	WT
P_1	0	12	30	30	18
P_2	2	6	18	16	10
P_3	4	8	28	24	16
P_4	6	4	20	14	10.

P1	P2	P1	P2	P3	P4	P1	P2	P2	P4	P1	P2	P1	P2	P1	
0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30.

avg tat = 21 units
avg wt = 13.5 .units

Task 6: Priority Scheduling

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	7	3	7	7	0
P_2	1	4	1	22	21	17
P_3	2	6	2	18	16	10
P_4	3	5	4	12	9	5

Grant Chart.

P_1	P_4	P_3	P_2	

t=0   7   12   18   22

avg TAT = $\frac{7+21+16+9}{4} = \frac{53}{4} = 12.25$ unit   avg WT = $\frac{17+10+5+0}{4} = \frac{32}{4} = 8$ unit

Task 7 :- Preemptive

Process	AT	BT	Priority	CT	TAT	WT
P_1	0	8	2	13	13	5
P_2	1	3	1	14	13	10
P_3	2	5	2	7	5	0

Grant Chart:

P_1	P_1	P_3	P_1	P_2

t=0   1   2   7   13   14

$$\text{avg TAT} = \frac{13+13+5}{3} = \frac{31}{3} = 10.333 \text{ unit.}$$

$$\text{avg WT} = \frac{5+10+0}{3} = 5 \text{ unit.}$$